

Linear Programming

Linear Programming

1. Linear programming handles any problem that can be formulated as an optimization over a set of variables where the goal (objective function) and constraints can all be expressed as linear functions of the variables
 - Applications to optimization research
2. Outline
 - Introduction
 - Examples
 - General Formulation
 - Simplex
 - Duality
 - Max-SAT approximation

Linear Programming Lecture Outline

1. Examples
 - Max-flow
 - Simple production
2. Standard form
3. LP duality

Max-Flow as LP

1. Input: Directed $G=(V,E)$ with capacities $c_e > 0$ for all edges
2. LP: m variables: f_e for every edge in E
 - Objective function: Max sum over flow out of a vertex
 - Subject to:
 - For every edge: $0 \leq f_e \leq c_e$
 - For every vertex: sum of flow in equals sum of flow out

Simple 2D Example

1. Basic production:
 - Company makes A and B
 - How many of each to maximize profit?
 - Each unit of A profits \$1 and B profits \$6
 - Demand: ≤ 300 units of A and ≤ 200 units of B
 - Supply: ≤ 700 hours, A takes 1 hour and B takes 3 hours

2D LP Formulation

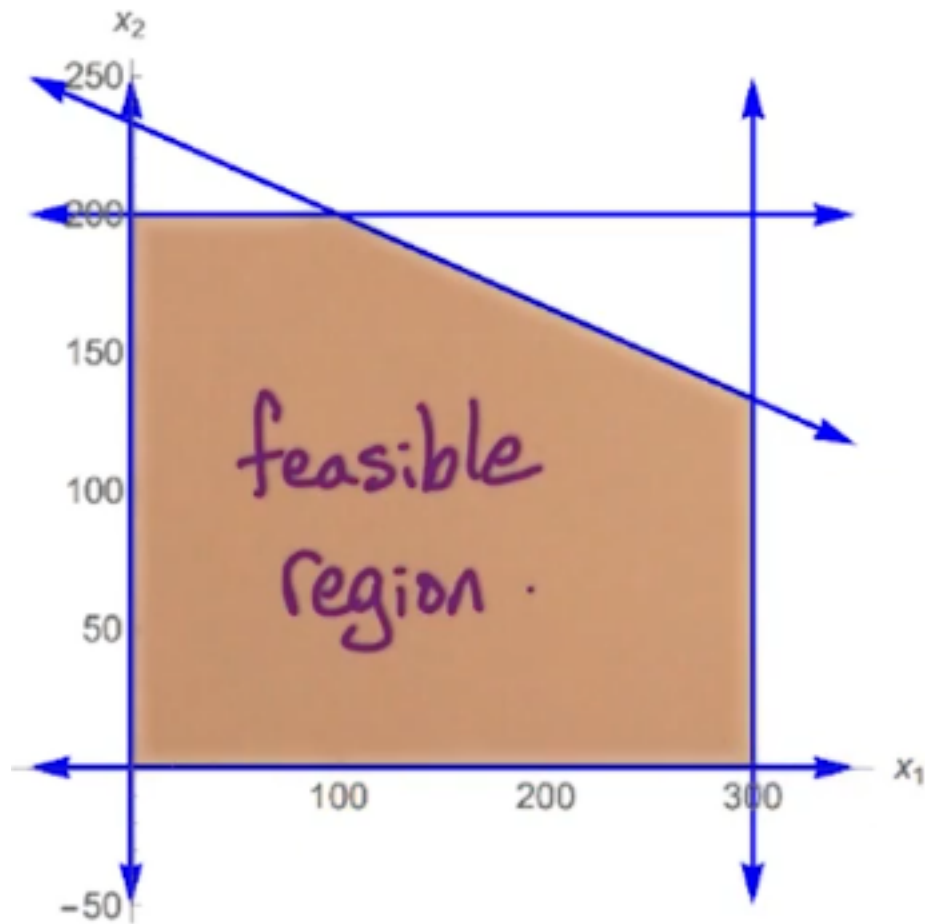
1. Variables
 - Let x_1 = # of units of A to produce per day
 - Let x_2 = # of units of B to produce per day
2. Objective function
 - $\max(x_1 + 6x_2)$
3. Constraints
 - $0 \leq x_1 \leq 300$
 - $0 \leq x_2 \leq 200$
 - $x_1 + 3x_2 \leq 700$

2D LP Recap

1. Formulation

- $\max(x_1 + 6x_2)$ such that
 - $x_1 \leq 300$
 - $x_2 \leq 200$
 - $x_1 \geq 0$
 - $x_2 \geq 0$
 - $x_1 + 3x_2 \leq 700$

2D Geometric View



2D Geometric View

Optimum

1. Goal: Maximize $x_1 + 6x_2 = c$
 - Solution:
 - $x_1 = 100$
 - $x_2 = 200$
 - $c = 1300$

Key Issues

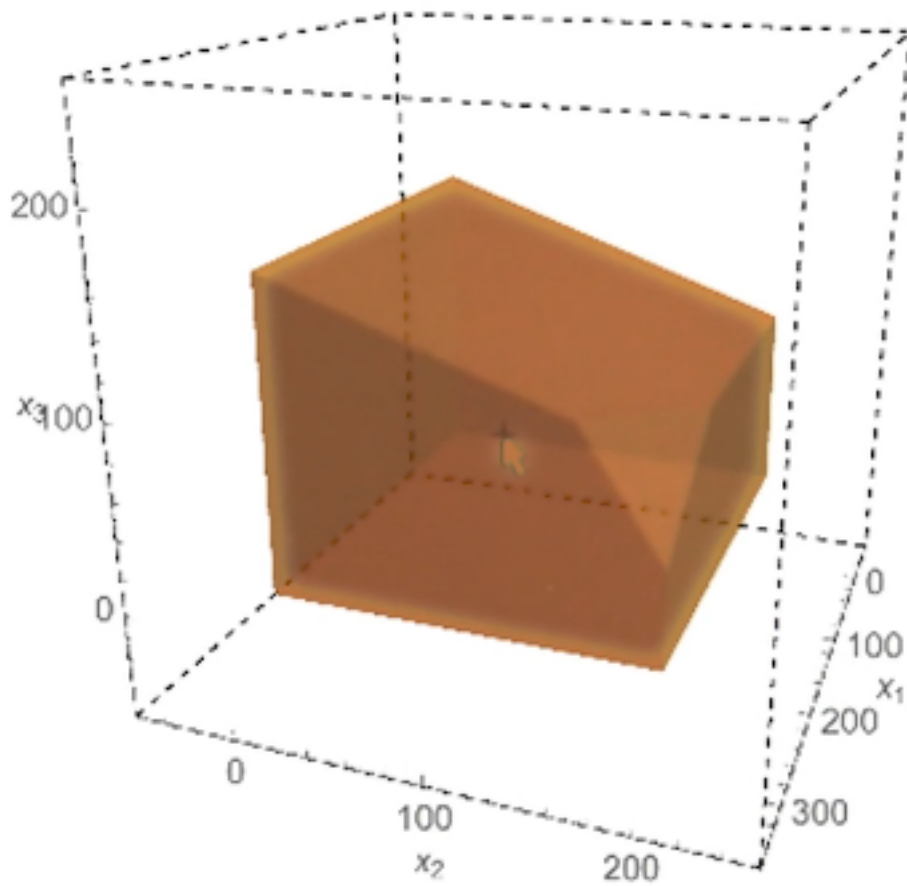
1. Optimum may be non-integer
 - What do we do if we require an integer solution? Can round
2. LP is in P
3. Integer linear programming (ILP) is NP-complete
4. Vertex = corner
5. Feasible region is convex
 - Any two points within the feasible region can be connected by a line totally contained within the set
 - Therefore, optimal point lies at a vertex
6. Simplex algorithm
 - Local greed approach
 - Start at a vertex and look at its neighboring vertices
 - Continue until we find a vertex that is better than all its neighbors

3D Example

1. Products A, B, C
 - Profit: \$1 for A, \$6 for B, \$10 for C
 - Demand: ≤ 300 for A, ≤ 200 for B, unlimited for C
 - Supply: ≤ 1000 total, A takes 1, B takes 3, C takes 2
 - Packaging: ≤ 500 total, A takes 0, B takes 1, C takes 3

3D LP Formulation

1. Objective: $\max(x_1 + 6x_2 + 10x_3)$
2. Constraints
 - Demand: $0 \leq x_1 \leq 300$
 - Demand: $0 \leq x_2 \leq 200$
 - Supply: $x_1 + 3x_2 + 2x_3 \leq 1000$
 - Packaging: $x_2 + 3x_3 \leq 500$
 - $x_1, x_2, x_3 \geq 0$



3D Geometric View

3D Geometric View

1. Standard form
 - n variables $x_1, \dots, x_n$
 - Objective function:
 - $\max(c_1x_1 + \dots + c_nx_n)$
 - Constraints:
 - $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$
 - $a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m$
 - $x_1, \dots, x_n \geq 0$

Standard Form

Linear Algebra View

variables $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

objective function $c = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$

constraint matrix

$m \times n$ A
constraints $b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$

$$\begin{array}{ll} \max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{array}$$

Linear Algebra

Converting to Standard Form

1. To convert from minimum to maximum, multiply by -1
 - $\min(c^T x) \Leftrightarrow \max(-c^T x)$
 - $a_1 x_1 + \dots + a_n x_n \geq b \Leftrightarrow -a_1 x_1 - \dots - a_n x_n \leq -b$
 - $a_1 x_1 + \dots + a_n x_n = b \Leftrightarrow a_1 x_1 + \dots + a_n x_n \leq b, \geq b$
2. Strict inequalities ($<$, $>$) are not allowed in linear programming
3. Unconstrained variable x (can be positive or negative)
 - Add x^+ and x^- where $x^+ \geq 0$, $x^- \geq 0$
 - $x = x^+ - x^-$

General Geometric View

1. n variables $\rightarrow n$ dimensions
 - $n + m$ constraints
 - Feasible region = Intersection of $n+m$ halfspaces = convex polyhedron

Vertices

1. Vertex of convex polyhedron:
 - Points satisfying n constraints with equality
 - Points satisfying m constraints with $\leq$
 - nm neighbors per vertex

LP Algorithms

1. Polynomial-time algorithms:

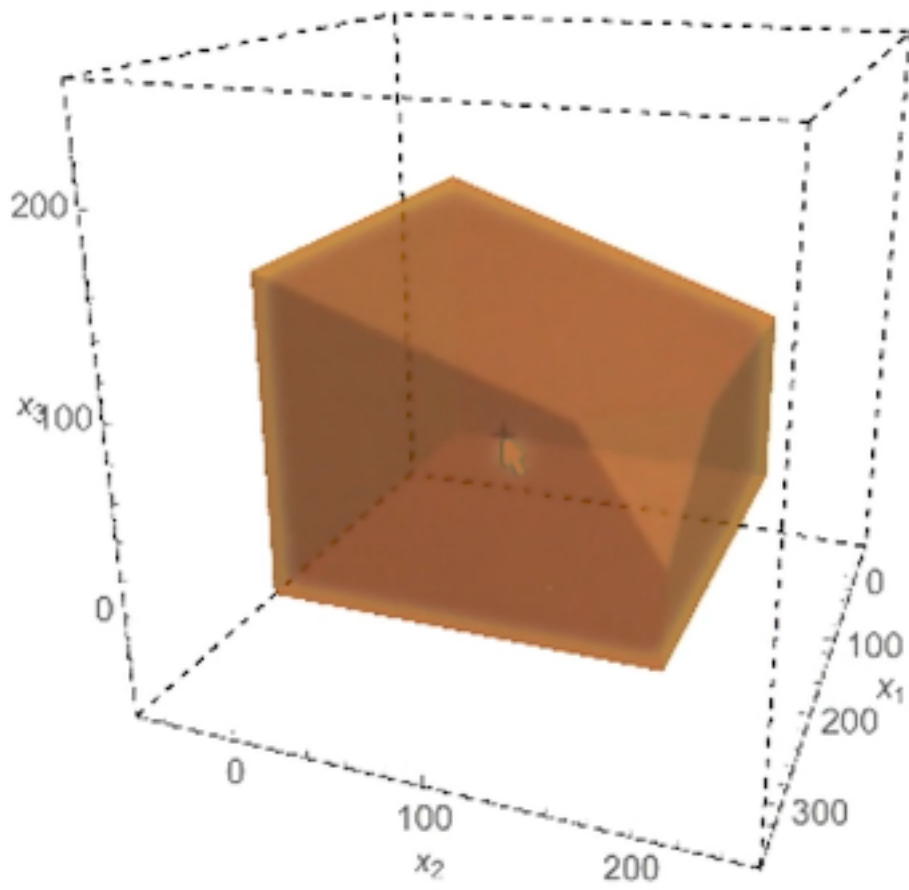
- Ellipsoid algorithms
 - Interior point methods
2. Simplex algorithm:
 - Worst-case exponential time
 - Widely used on HUGE LPs; guaranteed to give an optimal answer

Simplex Algorithm

1. Simplex algorithm:
 - Start at $x = 0$
 - Look for neighboring vertex with higher objective value
 - Then, move there and repeat
 - If there are multiple vertices with higher objective value, can use various heuristics
 - Pick one at random, pick largest
 - Because the feasible region is convex, the surface can not decrease and then increase again
 - Else, output(x)

Simplex Example

1. Start at $(0,0,0)$: profit = 0
2. Move to $(300,0,0)$: profit = 300
3. Move to $(300,200,0)$: profit = 1500
4. Move to $(300,200,50)$: profit = 2000
5. Move to $(200,200,100)$: profit = 2400



3D Geometric View
